








# Statistics of Financial Markets

## Chapter 13: Machine learning



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# Acronyms used in this material (1/2)

- **AI**: Artificial Intelligence
- **ARCH**: AutoRegressive Conditional Heteroskedasticity
- **AUC**: Area Under the (ROC) Curve
- **BET**: Bucharest Exchange Trading (index)
- **BVB**: Bursa de Valori București (the Bucharest Stock Exchange)
- **CV**: Cross-Validation
- **DM**: Diebold–Mariano (test)
- **DSR**: Deflated Sharpe Ratio
- **EODHD**: EOD Historical Data (market-data provider)
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- **GB**: Gradient Boosting
- **GJR-GARCH**: Glosten–Jagannathan–Runkle GARCH
- **HAR**: Heterogeneous AutoRegressive (model)
- **HS**: Historical Simulation
- **IBM**: International Business Machines (company)
- **MA**: Moving Average
- **ML**: Machine Learning
- **MLP**: MultiLayer Perceptron
- **MSE**: Mean Squared Error
- **NN**: Neural Network (NN$k$: $k$ hidden layers)

# Acronyms used in this material (2/2)

- **NN-ARCH**: Neural-Network ARCH
- **NN1**: Neural Network with 1 hidden layer
- **NN3**: Neural Network with 3 hidden layers
- **NN4**: Neural Network with 4 hidden layers
- **OLS**: Ordinary Least Squares
- **OLS-3**: OLS with 3 predictors (size, book-to-market, momentum)
- **OMV**: Österreichische Mineralölverwaltung — Austrian Mineral Oil Administration (OMV group)
- **PSR**: Probabilistic Sharpe Ratio
- **QGB**: Quantile Gradient Boosting
- **QLIKE**: Quasi-LIKElihood (loss function)
- **QR**: Quantile Regression
- **RBF**: Radial Basis Function (network)
- **RF**: Random Forest
- **RMSE**: Root Mean Squared Error
- **ROC**: Receiver Operating Characteristic (curve)
- **RV**: Realised Variance
- **SFE**: Statistics of Financial Markets, Exercises (Quantlet collection of the textbook)
- **SHAP**: SHapley Additive exPlanations
- **SSE**: Sum of Squared Errors
- **US**: United States

## Today's Question and Route

- **Question**: can flexible prediction methods forecast financial risk and returns better than the models of Chapters 7–10, and how do we check it honestly?
  - **machine learning** (ML): algorithms that learn a prediction rule from data, judged by their accuracy on new data
- **Route** of the chapter
  - prediction against inference; overfitting, bias and variance; validation for time series without look-ahead bias
  - ridge and lasso; trees, random forest, gradient boosting; neural networks and the neural-network ARCH of the textbook
  - three applications: realised volatility, the sign of returns, VaR 1%; data snooping; a landmark paper

## Learning Outcomes

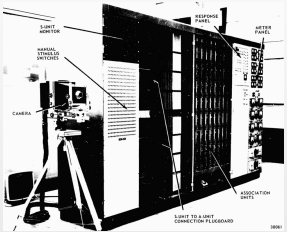
- Explain the difference between prediction and inference, and the bias–variance trade-off
- Design a walk-forward evaluation with purging, and recognise look-ahead bias and leakage
- Fit ridge, lasso, a regression tree, a random forest, gradient boosting and a small neural network
- Compare forecasts with out-of-sample $R^2$, QLIKE and the Diebold–Mariano test, against a proper baseline
- Forecast VaR 1% by quantile regression and correct a Sharpe ratio for the number of strategies tried

## Reading and Tools

- Textbook: Franke, Härdle & Hafner (2019), Ch. 19 (neural networks)
  - statistical learning in Python: James et al. (2023); in depth: Hastie, Tibshirani & Friedman (2009)
  - machine learning in finance: Kelly & Xiu (2023); López de Prado (2018)
- Python Quantlets of this chapter: Quantlets/Ch_13
  - ported from the SFE Quantlets SFEnnarch and SFEnnjpyusd; models from scikit-learn (small, fixed seeds, a few minutes in Colab)
- Lecture notebook: open in Google Colab
- Video courses: Machine learning in Financial Risk; Random Forests

# The Perceptron, 1958

- Rosenblatt (1958): a machine that learns to classify images by adjusting weights after each error
  - the **perceptron**: output 1 if $w_0 + w^\top x > 0$, else 0
  - $x$: the inputs (photocell signals); $w$: their weights, learned from examples; $w_0$: the threshold term; $w^\top x = \sum_j w_j x_j$
  - the Mark I Perceptron (Cornell, 1960): 400 photocells, weights set by electric motors
- Three later milestones
  - 1986: back-propagation trains networks with hidden layers Rumelhart, Hinton & Williams (1986)
  - 2001: random forests Breiman (2001b) and gradient boosting Friedman (2001)
  - 2024: Nobel Prize in Physics to Hopfield and Hinton for neural networks




The Mark I Perceptron, from its operator's manual

Figure: J.C. Hay, A.E. Murray (1960); public domain; Wikimedia Commons

# Two Cultures of Statistical Modelling

- Breiman (2001a), a statistician at Berkeley, describes two cultures
  - **data modelling**: assume a stochastic model (the linear regression $y = x^\top \beta + \varepsilon$, GARCH), estimate it, test its parameters
  - **algorithmic modelling**: treat the mechanism as unknown; find a rule $\hat{f}$ that predicts $y$ well on new data
- This course used the first culture in Chapters 2–11
  - this chapter adds the second, and asks when it pays off in finance



Evans Hall, University of California, Berkeley, home of the Department of Statistics

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## Prediction against Inference

- **Inference**: is $\beta_j \neq 0$? how large is the effect? (standard errors, tests, Chapter 6)
  - example: does the leverage effect exist (GJR-GARCH, Chapter 9)?
- **Prediction**: how close is $\hat{y}$ to $y$ on data the model has never seen?
  - example: next week's volatility; tomorrow's VaR 1%
  - the coefficients may be uninterpretable; what matters is the error on new data
- The two goals can conflict
  - a biased estimator (ridge, lasso) can predict better than the unbiased OLS
  - a model that predicts well does not prove a causal mechanism

## Supervised Learning: Notation and Loss (1/2)

- Data $(x_t, y_t)$, $t = 1, \ldots, n$
  - $t$: the index of the observation (a day, a week); $n$: the number of observations
  - $x_t \in \mathbb{R}^p$: the **features**, $p$ numbers known at time $t$ (past volatilities, past returns)
  - $y_t$: the **target**, the quantity we want to forecast (next week's volatility)
- Two types of problems
  - **regression**: $y$ continuous (volatility, return)
  - **classification**: $y$ a class (up/down, default; Chapter 12)

## Supervised Learning: Notation and Loss (2/2)

- Learning = choose $\hat{f}$ in a family $\mathcal{F}$ that minimises the average **loss** $\frac{1}{n}\sum_t L(y_t, f(x_t))$
  - $f(x_t)$: the forecast of a rule $f$ for observation $t$; $L(y, \hat{y})$: the cost of forecasting $\hat{y}$ when $y$ is observed
  - $\mathcal{F}$: the candidate rules (all linear regressions, all trees of depth 3); $\hat{f}$: the rule chosen from the data
- Losses used in this course
  - squared error $(y - \hat{y})^2$ (MSE: mean squared error); QLIKE for variances (Chapter 8)
  - quantile (pinball) loss for VaR (Chapter 10); log loss for probabilities (Chapter 12)
- A **hyperparameter** is chosen before the fit
  - examples: tree depth, penalty $\lambda$, number of neurons
  - the parameters (coefficients, weights) are learned by the fit

## Where ML Helps in Finance, and Where It Struggles

- **Helps**: targets with structure and enough data
  - volatility is persistent (Chapters 8–9): a lot to predict, nonlinear effects possible
  - cross-sections with thousands of stocks and many characteristics (the case study)
- **Struggles**: a low **signal-to-noise ratio**
  - daily returns are close to unpredictable (Chapter 7): a flexible model fits the noise
  - markets change (regimes, crises): the past is an imperfect training set
  - one price history: we cannot repeat the experiment, and the observations are dependent

## Training Error and Test Error

- **Training error**: the average loss on the data used for the fit; **test error**: on new data from the same process
  - the training error always falls as the model becomes more flexible
- **Overfitting**: the model learns the noise of the training sample
  - symptom: a small training error and a large test error
  - a tree with one leaf per observation has zero training error and predicts badly
- **Underfitting**: the model is too rigid to capture the signal
  - example: a constant forecast of volatility in a world with volatility clustering

## The Bias–Variance Decomposition (1/2)

- Model: $y = f(x) + \varepsilon$, $E[\varepsilon] = 0$, $\mathrm{Var}(\varepsilon) = \sigma^2$
  - $f$: the true, unknown relation between $x$ and $y$; $\varepsilon$: the noise, unpredictable from $x$, with variance $\sigma^2$
  - $\hat{f}$: the estimate of $f$ from a random training sample; another sample would give another $\hat{f}$
- Expected test error at a new point $x_0$, with $y_0 = f(x_0) + \varepsilon_0$:

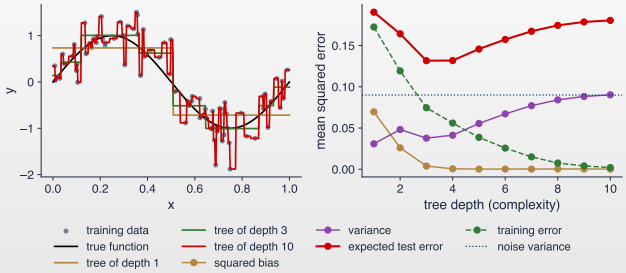
$$E\left[(y_0 - \hat{f}(x_0))^2\right] = \underbrace{\left(E[\hat{f}(x_0)] - f(x_0)\right)^2}_{\text{bias}^2} + \underbrace{\mathrm{Var}\left(\hat{f}(x_0)\right)}_{\text{variance}} + \underbrace{\sigma^2}_{\text{noise}}$$

  - $E[\cdot]$: the average over all possible training samples and over the noise $\varepsilon_0$
  - bias: how far the average estimate is from the truth; variance: how much $\hat{f}(x_0)$ moves from one sample to another

## The Bias–Variance Decomposition (2/2)

- Derivation: write $y_0 - \hat{f} = (f - E\hat{f}) + (E\hat{f} - \hat{f}) + \varepsilon_0$, all at $x_0$
  - square and take the expectation: the three squares give bias$^2$, variance and $\sigma^2$
  - the cross terms have mean 0: $E[E\hat{f} - \hat{f}] = 0$, and $\varepsilon_0$ is independent of the training sample
- Reading the decomposition
  - a rigid model: large bias, small variance
  - a flexible model: small bias, large variance
  - $\sigma^2$ is the floor: no model goes below it; in daily returns $\sigma^2$ is almost everything

# Bias and Variance of Regression Trees




- Simulation: $y = \sin(2\pi x) + \varepsilon$, $\sigma^2 = 0.09$, $n = 80$, 300 training samples; left: one sample with trees of depth 1, 3 and 10
- Depth 1: bias$^2$ 0.070, variance 0.031; depth 10: bias$^2$ 0.000, variance 0.090; the test error is smallest at depth 3 (0.132)

SFM_ch13_validation

## Interpretation: the U-Shaped Test Error

- The training error falls from 0.172 to 0.002; the test error falls, then rises
  - depth 10 has almost zero training error and a test error of 0.180, worse than depth 3
- Consequences for practice
  - never select a model by its training error
  - complexity is a hyperparameter: choose it on data that were not used for the fit
  - in finance the noise is large, so the best model is usually simpler than in this simulation

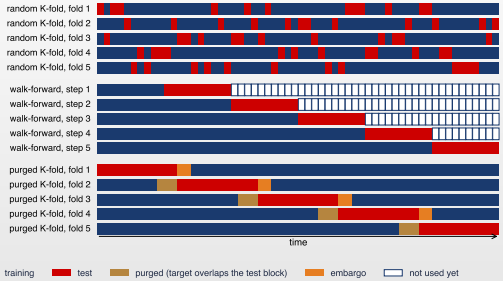
## Training, Validation and Test Samples

- Three roles for the data
  - **training**: estimate the parameters of each candidate model
  - **validation**: choose the hyperparameters and the model
  - **test**: measure the error of the final choice, once
- Every look at the test sample turns it into a validation sample
  - trying ten models on the test data and reporting the best is model selection, not testing
- **K-fold cross-validation**: split the data into $K$ folds; each fold is the validation set once, the others train the model
  - valid for i.i.d. data; for a time series only under strong conditions (no dependence left in the errors) Bergmeir, Hyndman & Koo (2018)

## Look-Ahead Bias and Leakage

- **Look-ahead bias**: the forecast for day $t$ uses information that was not available at the close of day $t-1$
  - features computed with future prices (a centred moving average, a full-sample mean)
  - scaling or standardising with statistics of the whole sample; revised data; today's index members (survivorship)
- **Leakage**: information about the test target reaches the training sample
  - overlapping targets: the 21-day return from day $t$ and from day $t+1$ share 20 days
  - a random fold puts day $t+1$ in training and day $t$ in test: the model has seen the answer
- Both make a model look better than it can be in real time

# Three Validation Schemes



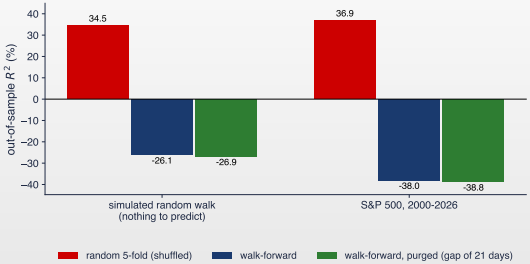

- **Walk-forward**: train on the past, test on the next block, move forward; the expanding window keeps all the past
- **Purged K-fold with embargo** López de Prado (2018): drop the training rows whose targets overlap the test block (purging) and a few rows after it (embargo)

SFM_ch13_validation

## Walk-Forward Validation, Step by Step

- The design used in this chapter for the S&P 500 (data from 8 April 2008)
  - one fold per calendar year from 2013: 14 folds; the model is re-estimated each January
  - training window: all earlier days (expanding); test: every day of the year
- Purging: the target is the mean variance of the next 5 days
  - for 2013 the last 5 training rows (from 24 December 2012) are dropped: their targets reach into 2013
  - the 2013 fold trains on 1 188 days and tests on 252; the 2026 fold trains on 4 458 days
- Inside each training window, hyperparameters are tuned by a walk-forward split of the training data only
  - no embargo is needed: the test block always comes after the training block

# Leakage in Practice: Random Folds on Overlapping Targets




- Random forest for the next 21-day return from past returns and volatilities; out-of-sample $R^2$ against the training mean
- Simulated random walk: random 5-fold 34.5%, walk-forward −26.1%; S&P 500 (6 631 days): 36.9% and −38.0%

SFM_ch13_validation

## Interpretation: a Forecast of Pure Noise with $R^2 > 30\%$

- The random walk has nothing to predict, yet random folds report $R^2 = 34.5\%$
  - the forest memorises neighbouring days, whose targets share 20 of 21 returns with the test day
- Walk-forward gives a negative $R^2$: the forest fits noise and loses against the training mean
  - purging (a gap of 21 days) changes little here, because in walk-forward only the boundary rows overlap
- Rule: with overlapping targets or dependent data, never shuffle the time order

## Many Features, Unstable OLS

- OLS (ordinary least squares): $\hat{\beta} = \arg\min_\beta \sum_t (y_t - x_t^\top \beta)^2 = (X^\top X)^{-1} X^\top y$
  - $X$: the $n \times p$ matrix whose row $t$ is $x_t^\top$; $y$: the vector of targets; $\arg\min_\beta$: the $\beta$ that minimises the sum
  - unbiased, but its variance $\sigma^2 (X^\top X)^{-1}$ ($\sigma^2$: the error variance) explodes when features are many or strongly correlated
- Volatility features are highly correlated: today's, this week's and this month's realised variance
  - OLS may give large coefficients of opposite signs that cancel in sample and fail out of sample
- **Penalised regression**: minimise the squared error plus a penalty on the size of $\beta$
  - accept a little bias for a large reduction in variance (the trade-off of Section 2)

## Ridge Regression (1/2)

- Hoerl & Kennard (1970): minimise the squared error plus a penalty on the squared coefficients

$$\hat{\beta}^{ridge} = \arg\min_{\beta} \sum_t (y_t - x_t^\top \beta)^2 + \lambda \sum_j \beta_j^2 = (X^\top X + \lambda I)^{-1} X^\top y$$

- Notation
  - $\beta_j$: the coefficient of feature $j$; $\sum_j \beta_j^2$: the penalty term, large when the coefficients are large
  - $\lambda \geq 0$: the strength of the penalty; $\lambda = 0$ gives OLS, $\lambda \to \infty$ gives $\hat{\beta} \to 0$
  - $I$: the $p \times p$ identity matrix
- Consequence
  - $X^\top X + \lambda I$ is always invertible for $\lambda > 0$: ridge works even with more features than observations

## Ridge Regression (2/2)

- **One feature**, centred (mean 0): $S_{xx} = \sum_t x_t^2$, $S_{xy} = \sum_t x_t y_t$

$$\hat{\beta}^{\,ridge} = \frac{S_{xy}}{S_{xx} + \lambda} = \frac{S_{xx}}{S_{xx} + \lambda}\,\hat{\beta}^{\,OLS}, \qquad \hat{\beta}^{\,OLS} = \frac{S_{xy}}{S_{xx}}$$

  - the factor $S_{xx}/(S_{xx} + \lambda)$ lies in $(0, 1]$: the share of the OLS coefficient that ridge keeps
- Example: $S_{xx} = 100$, $S_{xy} = 30$
  - OLS: $30/100 = 0.30$
  - ridge with $\lambda = 50$: $30/150 = 0.20$ (shrinkage factor 0.67)
- Ridge shrinks all coefficients towards zero, but sets none exactly to zero

# The Lasso

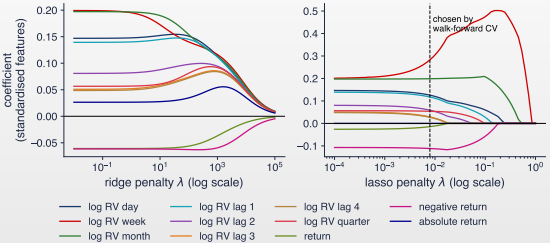
- Tibshirani (1996): $\hat{\beta}^{lasso} = \arg\min_\beta \sum_t (y_t - x_t^\top \beta)^2 + \lambda \sum_j |\beta_j|$
  - the absolute value has a kink at 0: some coefficients become **exactly zero** (variable selection)
- One feature: **soft thresholding**
  - $\hat{\beta}^{lasso} = \text{sign}(S_{xy})\ \max(|S_{xy}| - \lambda/2,\ 0)/S_{xx}$
  - $\text{sign}(S_{xy})$: $+1$ or $-1$; if $|S_{xy}| \leq \lambda/2$, the coefficient is exactly 0
  - same data, $\lambda = 20$: $(30 - 10)/100 = 0.20$; zero for $\lambda \geq 60$
- In practice
  - standardise the features first (the penalty depends on their units), with the training mean and standard deviation only
  - choose $\lambda$ by walk-forward cross-validation; elastic net combines both penalties



Robert Tibshirani, Stanford University

Photo: R. Tibshirani (2012); CC BY-SA 3.0; Wikimedia Commons

# Ridge and Lasso Paths for the Volatility Features




- Target: log of the mean variance proxy of the next 5 days, S&P 500, 2008–2012 (1 193 days); 11 standardised features (Section 7)
- Lasso with $\lambda = 0.0079$ (walk-forward CV): 10 of 11 features kept; the week coefficient rises from $+0.199$ (OLS) to $+0.283$, the absolute return drops out
- The design is exactly collinear, $|r| = r - 2\min(r, 0)$ (rank 10): the OLS coefficients of $r$, $\min(r, 0)$, $|r|$ are not unique (shown: the minimum-norm solution); the others are unique

SFM_ch13_shrinkage_trees

## Interpretation of the Paths

- Ridge (left): all coefficients shrink smoothly and together; correlated features share the weight
  - the negative return keeps a negative sign until the end: falling markets raise future volatility (the leverage effect, Chapter 9)
- Lasso (right): the features enter one by one as $\lambda$ falls
  - the weekly variance enters first: the most informative single feature
  - at the chosen $\lambda$ the lasso keeps the HAR structure (day, week, month) plus the negative return, $-0.112$
- The lasso selects; it does not test: a zero coefficient is not a p-value

## Regression Trees

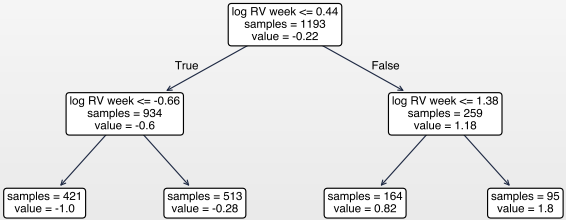
- A **regression tree** splits the feature space into rectangles and predicts the mean of $y$ in each **leaf**
  - split rule: a feature $j$ and a threshold $s$: left $x_j \leq s$, right $x_j > s$
- **Greedy fit**: at each node choose $(j, s)$ that minimises $\mathrm{SSE} = \sum_{\text{left}}(y - \bar{y}_L)^2 + \sum_{\text{right}}(y - \bar{y}_R)^2$
  - SSE: sum of squared errors; $\bar{y}_L$, $\bar{y}_R$: the means of $y$ in the left and right groups (the two forecasts)
  - repeat in each half until a stopping rule: maximum depth, minimum number of observations per leaf
- Strengths and weaknesses
  - nonlinear effects and interactions without specifying them; insensitive to monotone transformations of $x$
  - high variance: a small change in the data can change the whole tree (Section 2)

# Worked Example: One Split, Step by Step

| $s$ | $\bar{y}_L$ | $\bar{y}_R$ | SSE |
|---|---|---|---|
| 0.70 | 0.70 | 1.52 | 1.708 |
| 0.90 | 0.80 | 1.68 | 1.248 |
| 1.25 | 0.80 | 1.97 | 0.227 |
| 1.75 | 1.00 | 2.15 | 0.505 |
| 2.20 | 1.22 | 2.20 | 1.468 |

- Six weeks: $x$ = this week's volatility (%), $y$ = next week's
  - $x$: 0.6, 0.8, 1.0, 1.5, 2.0, 2.4; $y$: 0.7, 0.9, 0.8, 1.6, 2.1, 2.2
  - no split: $\bar{y} = 1.38$, SSE $= 2.268$
- Candidates: the midpoints between consecutive $x$ values
  - best split $s = 1.25$: leaves 0.80 and 1.97, SSE $= 0.227$
  - one split explains $1 - 0.227/2.268 = 0.90$ of the variation

# A Depth-2 Tree for Next Week's Volatility




- S&P 500, 2008–2012, target and features in log variance ($\%^2$ per day); *value*: the mean of the target in the node
- Interpretation: the first split, log RV week $\leq 0.44$, is an annual volatility of about 19.8%; all splits use the weekly variance, the feature the lasso picked first

SFM_ch13_shrinkage_trees

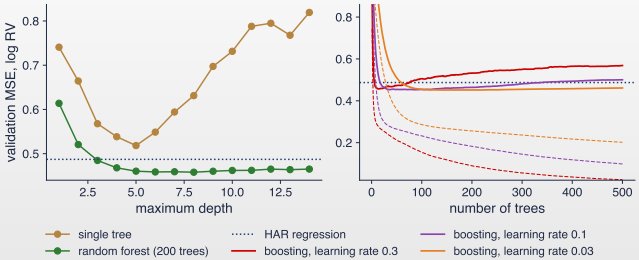
## Random Forest

- Breiman (2001b): average $B$ deep trees, each grown on a bootstrap sample (Chapter 2) of the training data (**bagging**)
  - at each split only a random subset of the features is tried: the trees become less correlated
- Why averaging works: $B$ trees, each with variance $\sigma^2$ and pairwise correlation $\rho$
  - $\mathrm{Var}(\text{average}) = \rho\sigma^2 + (1-\rho)\sigma^2/B$: the part $\rho\sigma^2$, common to all trees, does not fall with $B$
  - with $\rho = 0.3$, $\sigma^2 = 1$: $B = 1$: 1.000; $B = 10$: 0.370; $B = 300$: 0.302
  - more trees never overfit; less correlated trees (fewer features per split) reduce the floor $\rho\sigma^2$
- Hyperparameters used here: 300 trees, at least 20 observations per leaf, half of the features tried at each split

## Gradient Boosting

- Friedman (2001): build the model step by step, each new small tree fitting what the previous ones missed
  - start with $F_0 = \bar{y}$; at step $m$, fit a shallow tree $h_m$ to the residuals $y_t - F_{m-1}(x_t)$
  - update $F_m = F_{m-1} + \nu\, h_m$, with the **learning rate** $0 < \nu \leq 1$
  - for squared loss the residuals are minus the gradient of the loss: hence the name
- Random forest against boosting
  - forest: deep trees in parallel, averaging reduces variance
  - boosting: shallow trees in sequence, each step reduces bias; too many steps overfit
- Any differentiable loss works: quantile loss gives quantile boosting (Section 9)

# One Tree, a Forest and Boosting on Volatility Data




- S&P 500, training 2008–2018 (2 698 days), validation 2019–2022 (1 008 days); dashed right: training MSE; dotted: HAR regression (0.487)
- Single tree: best 0.518 at depth 5, 0.819 at depth 14; forest: 0.458 (depth 8) and still 0.465 at depth 14

SFM_ch13_shrinkage_trees

## Interpretation: Averaging and Early Stopping

- The single tree overfits quickly; the forest is almost insensitive to depth
  - averaging removes most of the variance of deep trees; the forest beats HAR on this validation period
- Boosting: the validation error reaches its minimum, then rises while the training error keeps falling
  - learning rate 0.3: minimum 0.457 after 13 trees, 0.568 after 500
  - learning rate 0.03: minimum 0.451 after 154 trees, 0.461 after 500: smaller steps are safer
- The number of trees of boosting is a hyperparameter; the number of trees of a forest is not

## From the Perceptron to the Multilayer Perceptron (1/2)

- A **neuron**: $h = g(b + w^\top x)$, a weighted sum of the inputs passed through an **activation function** $g$
  - $x$: the inputs; $w$: their weights; $b$: a constant (the bias term of the neuron); $h$: the output of the neuron
  - $g$: a fixed nonlinear function; without it, a network of neurons would be just a linear regression
- Activation functions
  - the perceptron uses a step function (0 or 1)
  - modern networks use smooth or piecewise linear $g$: logistic $1/(1 + e^{-z})$, tanh, ReLU $\max(0, z)$

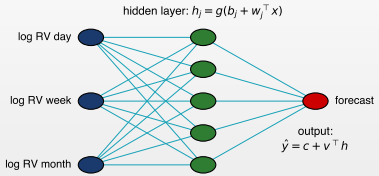
# From the Perceptron to the Multilayer Perceptron (2/2)

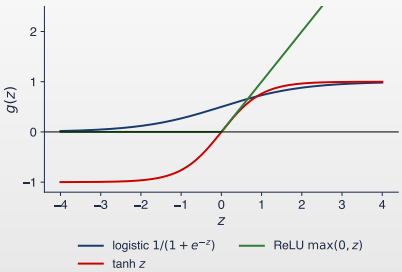
- **MLP** (multilayer perceptron) with one hidden layer of $K$ neurons

$$\hat{y} = c + \sum_{k=1}^{K} v_k \, g(b_k + w_k^\top x)$$

  - $w_k$, $b_k$: the weights and the constant of hidden neuron $k$; $v_k$: the weight of neuron $k$ in the output; $c$: the output constant
  - reading: a linear model in $K$ nonlinear features $g(b_k + w_k^\top x)$, which are themselves learned
- Number of parameters with 11 inputs and 16 neurons
  - hidden layer: $11 \times 16$ weights $+16$ constants $= 192$; output: $16 + 1 = 17$; total 209
- **Universal approximation** Hornik, Stinchcombe & White (1989): with enough neurons, one hidden layer approximates any continuous function on a compact set

# A Small Network and Its Activation Functions




- Left: three HAR inputs, five hidden neurons, one forecast; each line is a weight
- Right: logistic and tanh saturate for large $|z|$; ReLU $= \max(0, z)$ is linear for $z > 0$ and trains faster

SFM_ch13_neural_networks

## Training a Network

- Minimise the loss by **gradient descent**: $\theta \leftarrow \theta - \eta \nabla_\theta L$
  - $\theta$: all weights and constants; $\nabla_\theta L$: the gradient, the vector of partial derivatives of the loss; $\eta > 0$: the learning rate (the step size)
  - each step moves $\theta$ in the direction in which the loss falls fastest
  - **back-propagation** computes $\nabla_\theta L$ layer by layer with the chain rule Rumelhart, Hinton & Williams (1986)
  - one **epoch** = one pass over the training data; Adam is a gradient method with adaptive step sizes
- The loss is not convex: different starting weights give different networks
  - we fix the random seeds and average five networks (an **ensemble**)
- Against overfitting
  - standardise inputs and target with training statistics; an $L_2$ penalty on the weights (as ridge); a fixed number of epochs
  - small data, small network: one hidden layer of 16 neurons is enough here

# From Deep Blue to Deep Learning



IBM Deep Blue, Computer History Museum

Photo: Anton Chiang (2011); CC BY 2.0; Wikimedia Commons



Geoffrey Hinton, Nobel Prize in Physics 2024

Photo: Arthur Petron (2024); CC BY-SA 4.0; Wikimedia Commons

- 1997: Deep Blue beat the world chess champion mainly by search and hand-crafted evaluation rules, not by learning from data
- Deep learning (many layers, huge data) now dominates images and text; on financial prices the data are few and noisy, so small networks are the norm

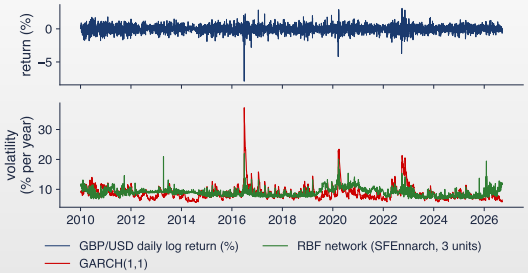
# Neural-Network ARCH: the Textbook Model (1/2)

- Franke, Härdle & Hafner (2019), Ch. 19: $r_t = f(x_{t-1}) + s(x_{t-1})\,\varepsilon_t$, with $f$ and $s^2$ estimated by neural networks
  - $r_t$: the return of day $t$; $x_{t-1}$: the inputs known at the close of day $t-1$
  - $f(x_{t-1})$: the conditional mean; $s^2(x_{t-1})$: the conditional variance; $\varepsilon_t$: a shock with mean 0 and variance 1
- ARCH (Engle (1982)) is the special case $s^2(x) = \omega + \sum_i \alpha_i r_{t-i}^2$
  - $\omega > 0$, $\alpha_i \geq 0$: the variance is a linear function of the past squared returns
  - the network imposes no shape on $s^2$

# Neural-Network ARCH: the Textbook Model (2/2)

- The Quantlet SFEnnarch: GBP/USD, inputs $x$ = the last 3 returns, the German 10-year yield, the gold return
  - step 1: a network $\hat{f}$ for the mean; residuals $e_t = r_t - \hat{f}(x_{t-1})$
  - step 2: a network for $e_t^2$ gives the conditional variance $\hat{s}^2(x_{t-1})$
- **RBF** (radial basis function) network with 3 units, as in the Quantlet
  - unit $j$: $\exp(-\|x - c_j\|^2 / s_j^2)$; $c_j$: its centre (found by clustering); $s_j$: its width; $\|x - c_j\|$: the distance from $x$ to $c_j$
  - the unit is close to 1 when $x$ is near $c_j$ and close to 0 far from it; output: a linear combination of the 3 units
  - nothing forces $\hat{s}^2 > 0$: a variance network needs a floor, which GARCH guarantees by its constraints

# Neural-Network ARCH for GBP/USD




- Daily data from 7 January 2010 (4 048 days); annualised volatility of the RBF network and of GARCH(1,1) ($\hat{\alpha} = 0.077$, $\hat{\beta} = 0.892$)
- Average volatility: network 8.9%, GARCH 8.8%; correlation of the two paths 0.45 (a small MLP: 0.43)

SFM_ch13_neural_networks

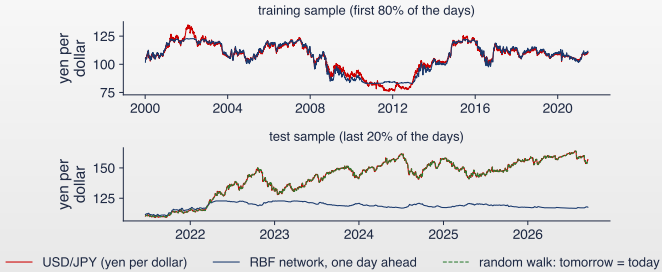
## Interpretation: Flexible Shape, Short Memory

- After the Brexit vote, GARCH peaks at 37.3% (28 June 2016); the network only at 22.0%
  - the network sees only the last 3 returns: one large return raises its variance for 3 days, then the effect is gone
  - GARCH(1,1) (Chapter 9): $\sigma_t^2 = \omega + \alpha\, r_{t-1}^2 + \beta\, \sigma_{t-1}^2$, with $\sigma_{t-1}^2$ yesterday's conditional variance
  - the term $\beta\, \sigma_{t-1}^2$ carries the whole past: volatility clustering needs memory, not only flexibility
- The mean network explains 0.08% of the returns in sample: exchange rates are close to a random walk
  - the useful part of NN-ARCH is the variance, as for GARCH
- Lesson: good features (HAR averages, Section 7) matter more than the type of network

## Forecasting an Exchange Rate Level: SFEnnjpyusd

- The Quantlet SFEnnjpyusd forecasts the yen–dollar rate from its last 3 values with an RBF network of 3 units
  - first 80% of the days for training, last 20% for testing; here USD/JPY from 2000, split on 21 June 2021
- Two traps of level forecasting
  - a level series looks predictable because $P_t \approx P_{t-1}$: the right benchmark is the random walk $\hat{P}_t = P_{t-1}$ (Chapter 7)
  - a network with bounded units cannot extrapolate: the test levels go up to 163.9, the training maximum was 134.9
- The original code rescales the test forecasts with the range of the test series, which is unknown in real time; here the training range is used

# The RBF Network against the Random Walk




- RMSE (root mean squared error) in yen: training: network 2.64, random walk 0.66; test: network 27.56, random walk 0.89
- Interpretation: out of sample the network is 31 times worse than "tomorrow = today"; on returns the same network has an out-of-sample $R^2$ of $-0.14\%$

SFM_ch13_neural_networks

## Target and Features (1/2)

- Daily variance proxy of Chapter 8 (no intraday data): an estimate of the variance of day $t$ from its open, high, low and close

$$v_t = o_t^2 + \tfrac{1}{2}(u_t - d_t)^2 - (2\ln 2 - 1)c_t^2$$

  - $o_t$: the overnight log return (from the previous close to the open)
  - $u_t, d_t, c_t$: log high, low and close relative to the open (the Garman–Klass term, Garman & Klass (1980))

- **Target**: $y_t = \ln\left(\frac{1}{5}\sum_{j=1}^{5} v_{t+j}\right)$, the log realised variance of the next week
  - $j = 1, \ldots, 5$: the next five trading days; the target is known only at the close of day $t + 5$
  - logs make the target closer to Normal and the forecasts positive after exp

## Target and Features (2/2)

- **Features** at the close of day $t$
  - HAR: $\ln v_t$, ln of the mean over 5 days (week) and over 22 days (month)
  - extended (11): plus 4 lags of $\ln v_t$ ($\ln v_{t-1}, \ldots, \ln v_{t-4}$), the 66-day mean, $r_t$, $\min(r_t, 0)$, $|r_t|$
- Notation
  - $r_t$: the log return of day $t$; $|r_t|$: its absolute value
  - $\min(r_t, 0)$: the return if negative, 0 otherwise; it lets falls and rises act differently (the leverage effect)

# The HAR Benchmark (1/2)

- **HAR** (heterogeneous autoregressive) model Corsi (2009)

$$y_t = \beta_0 + \beta_d \ln v_t + \beta_w \ln v_t^{(w)} + \beta_m \ln v_t^{(m)} + u_t$$

  - $v_t^{(w)}$, $v_t^{(m)}$: the mean of $v$ over the last 5 days (week) and over the last 22 days (month)
  - $\beta_0$: the constant; $\beta_d, \beta_w, \beta_m$: the weights of the three horizons; $u_t$: the error

- Interpretation
  - three horizons of traders (daily, weekly, monthly); estimated by OLS
  - with positive weights on the three horizons, a shock to volatility fades slowly
  - a simple linear model that mimics long memory (Chapter 11) and is hard to beat

## The HAR Benchmark (2/2)

- The question of this section
  - do lasso, random forest, gradient boosting or an MLP add anything to HAR out of sample?
  - a large study with intraday data finds modest gains, mainly from extra features Christensen, Siggaard & Veliyev (2023)
- Design: walk-forward of Section 3, yearly refits; S&P 500 from 2013, Bitcoin from 2018
  - variance forecast $\hat{h}_t = \exp(\hat{y}_t + \hat{s}^2/2)$, $\hat{s}^2$ the residual variance in training (the log-Normal mean)
  - $\hat{y}_t$: the forecast of the log variance; $\exp(\hat{y}_t)$ alone would underestimate the mean variance, hence the term $\hat{s}^2/2$

# Evaluation (1/3): Out-of-Sample $R^2$

- **Out-of-sample** $R^2$ against a benchmark forecast $\tilde{y}_t$ Campbell & Thompson (2008)

$$R^2_{OOS} = 1 - \frac{\sum_t (y_t - \hat{y}_t)^2}{\sum_t (y_t - \tilde{y}_t)^2}$$

  - $\hat{y}_t$: the forecast of the model; sums over the test days only
  - benchmark $\tilde{y}_t$: the training mean, or HAR

- Reading
  - $R^2_{OOS} \leq 1$; $R^2_{OOS} > 0$: smaller squared error than the benchmark; $R^2_{OOS} < 0$: worse than the benchmark
  - unlike the in-sample $R^2$, it can be negative

- Step by step, S&P 500, MLP against HAR
  - $1 - 0.397/0.433 = 1 - 0.917$, i.e. $+8.3\%$

# Evaluation (2/3): QLIKE

- **QLIKE** (Chapter 8): a loss for variance forecasts, robust to a noisy proxy Patton (2011)

$$L_t = \frac{\mathrm{RV}_t}{\hat{h}_t} + \ln \hat{h}_t$$

  - $\mathrm{RV}_t$: the realised variance of the next 5 days (the target, in levels); $\hat{h}_t$: its forecast
  - for a given $\mathrm{RV}_t$, the loss is smallest at $\hat{h}_t = \mathrm{RV}_t$; lower average QLIKE is better
- Asymmetry
  - for $\mathrm{RV} = 1$: $\hat{h} = 0.5$ costs 1.307, $\hat{h} = 2$ costs 1.193
  - under-prediction is punished more: for risk management, underestimating volatility is the costly error

# Evaluation (3/3): the Diebold–Mariano Test

- **Diebold–Mariano** test Diebold & Mariano (1995): do two models $A$ and $B$ have the same expected loss?

$$d_t = L_t^A - L_t^B, \qquad \mathrm{DM} = \frac{\bar{d}}{\sqrt{\widehat{\mathrm{LRV}}/n}} \sim N(0,1) \text{ under equal accuracy}$$

  - $L_t^A$, $L_t^B$: the losses of the two models on day $t$; $\bar{d}$: the mean of $d_t$; $n$: the number of test days
  - $\widehat{\mathrm{LRV}}$: the Newey–West long-run variance of $d_t$ (5 lags), because overlapping targets make $d_t$ autocorrelated
- Reading
  - $\mathrm{DM} < 0$: $A$ has the smaller loss; $\mathrm{DM} > 0$: $B$ is better
  - $|\mathrm{DM}| > 1.96$: the difference is significant at 5%

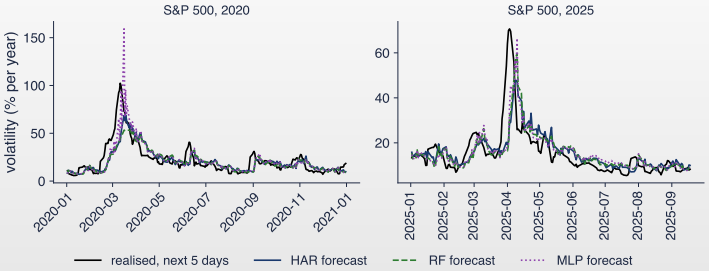
# Results: S&P 500 and Bitcoin

| | Model | $R^2_{OOS}$ vs. mean | $R^2_{OOS}$ vs. HAR | QLIKE | DM vs. HAR (p) |
|---|---|---|---|---|---|
| S&P 500 | HAR | 61.5 | – | 0.414 | – |
| | Lasso | 64.2 | +7.1 | 0.400 | −5.12 (< 0.001) |
| | RF | 61.9 | +1.0 | 0.416 | +0.32 (0.747) |
| | GB | 60.4 | −2.7 | 0.421 | +1.02 (0.308) |
| | MLP | 64.7 | +8.3 | 0.398 | −3.96 (< 0.001) |
| Bitcoin | HAR | 44.0 | – | 3.111 | – |
| | Lasso | 45.1 | +2.0 | 3.108 | −0.64 (0.520) |
| | RF | 43.4 | −1.2 | 3.119 | +0.81 (0.418) |
| | GB | 41.7 | −4.2 | 3.148 | +2.12 (0.034) |
| | MLP | 43.8 | −0.3 | 3.123 | +1.10 (0.272) |

- Forecasts for 3 444 days (S&P 500, from 2 January 2013) and 3 178 days (Bitcoin, from 1 January 2018); $R^2_{OOS}$ in %, on log RV; GB: gradient boosting
- S&P 500: lasso and MLP beat HAR significantly (+7.1%, +8.3%); random forest and boosting do not
- Bitcoin: no model beats HAR; boosting is significantly worse (DM +2.12)

SFM_ch13_volatility_forecast

# Forecasts in Two Storms: 2020 and 2025



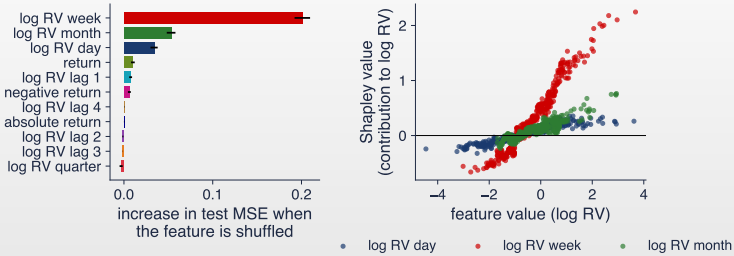

- S&P 500: annualised realised volatility of the next 5 days, $\sqrt{252\,\mathrm{RV}_t}$, and the HAR, random forest and MLP forecasts made at the close of day $t$
- Interpretation: all models react only after the first shock (March 2020, April 2025); the forest is capped by the largest values it saw in training, the MLP can overshoot

SFM_ch13_volatility_forecast

## Feature Importance: Permutation and Shapley Values

- **Permutation importance**: shuffle one feature in the test data and measure how much the test MSE rises
  - large increase: the model relies on the feature; correlated features share (and hide) their importance
- **Shapley values** (game theory): $\hat{f}(x) = \phi_0 + \sum_j \phi_j(x)$, $\phi_j$ the fair share of feature $j$ in the forecast
  - $\phi_0$: the average forecast; $\phi_j(x) > 0$: feature $j$ raises this forecast above the average, $\phi_j(x) < 0$: lowers it
  - $\phi_j$ averages the change of $\hat{f}$ when $j$ joins every subset of the other features
  - SHAP (Shapley additive explanations) Lundberg & Lee (2017): fast versions for trees; for a linear model $\phi_j = \beta_j(x_j - \bar{x}_j)$
- Both describe the model, not the economy: importance is not causality

# The Features Used by the Forest




- Left: random forest on 11 features, trained to 2019, tested 2020–2026 (1 682 days): shuffling the weekly variance raises the MSE by 0.201, the monthly by 0.053, the daily by 0.034
- Right: exact Shapley values of a forest on the 3 HAR features; mean $|\phi|$: week 0.52, month 0.13, day 0.13; the effect is nonlinear and grows with the level of volatility

SFM_ch13_volatility_forecast

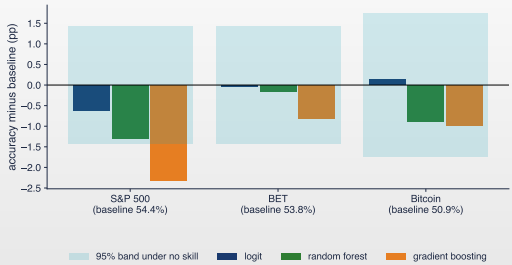
## A Classification Problem

- Target: $y_t = 1\{r_{t+1} > 0\}$; features at the close of day $t$: the last 5 returns, sums over 5, 21 and 63 days, volatility over 21 and 63 days
  - $1\{\cdot\}$: the indicator, 1 if the condition holds and 0 otherwise; $y_t = 1$: tomorrow's return is positive
  - models: logit (Chapter 12), random forest and gradient boosting classifiers; walk-forward, yearly refits
  - S&P 500 and BET from 2008, Bitcoin from 2018
- Metrics (defined in Chapter 12)
  - accuracy: the share of days with the right sign; AUC (area under the ROC curve): 0.5 means no ranking skill
- Why expect little: weak-form efficiency (Chapter 7)
  - any reliable pattern in past returns would be traded away

## The Right Baseline

- **Majority-class baseline**: always forecast the more frequent class of the training sample (here: "up")
  - S&P 500 since 2008: 54.4% of the days are up, so "always up" is right on 54.4% of the days
  - an accuracy of 53% sounds like skill; against this baseline it is a loss
- Test against the baseline: $z = (\widehat{\mathrm{acc}} - \mathrm{acc}_0)/\sqrt{\mathrm{acc}_0(1-\mathrm{acc}_0)/n}$
  - $\widehat{\mathrm{acc}}$: the accuracy of the model; $\mathrm{acc}_0$: that of the baseline; $n$: the test days; $z \approx N(0,1)$ if the model has no extra skill
  - S&P 500, $n = 4\,705$: standard error 0.73 points; a 95% band of $\pm 1.4$ points around the baseline
  - random forest: 53.1%, $z = -1.79$ (p 0.074); boosting: 52.1%, $z = -3.19$ (p 0.001)
- For returns, the matching baseline is the zero or historical-mean forecast Welch & Goyal (2008)

# Accuracy against the Baseline




- Accuracy minus the baseline accuracy (percentage points); band: $\pm 1.96$ standard errors around zero
- AUC of the logit and of the forest: S&P 500 0.511 and 0.507, BET 0.533 and 0.533, Bitcoin 0.515 and 0.519; logit forecasts "up" on 86% of S&P 500 days

SFM_ch13_sign_prediction

## Interpretation: Why Sign Prediction Mostly Fails

- No model beats "always up" significantly on any market; the flexible models do worse than the logit
  - S&P 500: logit $-0.6$, forest $-1.3$, boosting $-2.3$ points; Bitcoin: logit $+0.1$, forest $-0.9$
- The signal-to-noise ratio of daily returns is tiny
  - flexible models find patterns in the noise of the training years (Section 2); the logit is close to the baseline because it hardly moves
- Even a real edge of one point would rarely survive costs: a daily sign strategy trades very often
  - ML gains in returns come from large cross-sections at monthly horizons, not from one index tomorrow (Section 11)

## Quantile Regression (1/2)

- Koenker & Bassett (1978): estimate the conditional $\alpha$-quantile $q_\alpha(x) = x^\top \beta_\alpha$ by minimising the **quantile (pinball) loss**
  - $q_\alpha(x)$: the level below which $y$ falls with probability $\alpha$, given $x$; $\beta_\alpha$: coefficients specific to the level $\alpha$
- The loss of an error $u = y - q$:

$$\rho_\alpha(u) = u\,(\alpha - 1\{u < 0\})$$

  - below the quantile ($u < 0$) the error costs $(1-\alpha)|u|$, above it $\alpha|u|$
  - with $\alpha = 0.01$, the minimum is reached when about 1% of the observations lie below $q$

## Quantile Regression (2/2)

- With $\alpha = 0.01$ and $q = -\mathrm{VaR}$: the average quantile loss of Chapter 10
  - $(\alpha - I_t)(r_t + \mathrm{VaR}_t)$, with $I_t = 1$ if $r_t < -\mathrm{VaR}_t$ (an exception) and 0 otherwise
- VaR 1% for day $t+1$: $\widehat{\mathrm{VaR}}_t = -\hat{q}_{0.01}(x_t)$, with $x_t$ = daily, weekly and monthly volatility (square roots of the proxy) and $r_t$
  - no distribution is assumed: the data choose how volatility maps into the tail; CAViaR Engle & Manganelli (2004) is a dynamic version
- **Quantile gradient boosting**: the same loss with boosted trees (Section 5)
  - flexible, but at 1% each leaf sees few tail observations: high variance

# Backtest of VaR 1%: Quantile Methods, HS and GARCH-t (1/2)

| | Method | $x$ | rate (%) | p Kupiec | p ind. | loss $\times 100$ | mean VaR |
|---|---|---|---|---|---|---|---|
| S&P 500 | QR | 28 | 0.81 | 0.252 | 0.020 | 3.38 | 2.70 |
| | QGB | 42 | 1.22 | 0.213 | 0.015 | 3.51 | 2.46 |
| | HS | 46 | 1.33 | 0.061 | $<0.001$ | 4.78 | 2.97 |
| | GARCH-t | 52 | 1.51 | 0.005 | 0.008 | 3.50 | 2.42 |
| Bitcoin | QR | 23 | 0.72 | 0.098 | 0.563 | 13.03 | 10.46 |
| | QGB | 47 | 1.48 | 0.012 | 0.235 | 13.19 | 8.86 |
| | HS | 32 | 1.01 | 0.974 | 0.332 | 13.01 | 9.30 |
| | GARCH-t | 41 | 1.29 | 0.117 | 0.558 | 13.15 | 8.52 |

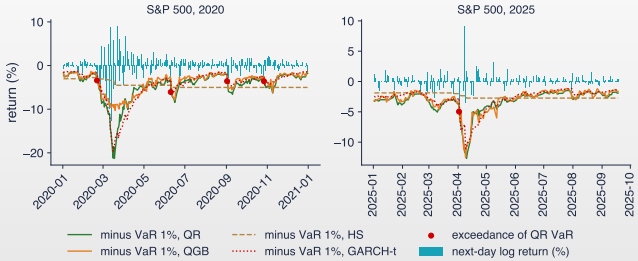
- Columns
  - $x$: the number of exceptions ($r_{t+1} < -\widehat{\mathrm{VaR}}_t$) in $n$ test days; rate $= x/n$, target 1%
  - p Kupiec: p-value of the Kupiec test of the rate; p ind.: p-value of the Christoffersen test of independence (Chapter 10)
  - loss $\times 100$: the mean quantile loss (smaller is better); mean VaR: the average VaR 1%, in %
- Methods
  - QR: linear quantile regression; QGB: quantile gradient boosting
  - HS: historical simulation on 500 days; GARCH-t: re-estimated each January (Chapter 10)

SFM_ch13_quantile_var

# Backtest of VaR 1%: Quantile Methods, HS and GARCH-t (2/2)

- S&P 500 (3 448 days from 2 January 2013): QR has the right rate (0.81%) and the smallest loss
  - its exceptions are not fully independent (p ind. 0.020)
  - GARCH-t is exceeded too often (1.51%)
- Bitcoin (3 182 days): HS has the best rate (1.01%)
  - QGB is rejected by the Kupiec test (1.48%, p 0.012)
- Reading the p-values
  - p Kupiec below 0.05: the exception rate differs significantly from 1%
  - p ind. below 0.05: the exceptions come in clusters, so the VaR reacts too slowly

# VaR 1% Forecasts in 2020 and 2025




- S&P 500: next-day returns and minus VaR 1% of the four methods; dots: exceedances of the quantile-regression VaR
- Interpretation: QR and GARCH-t move with volatility; HS reacts only slowly; in 2020 HS was exceeded 10 times, QR 4 times, and the HS exceptions cluster (p ind. $< 0.001$)

SFM_ch13_quantile_var

## The Best of $N$ Strategies (1/2)

- **Data snooping**: trying many models or rules on the same data and reporting the best White (2000)
  - with enough trials, a rule without skill looks excellent in sample
  - hundreds of published return predictors would not survive a multiple-testing correction Harvey, Liu & Zhu (2016)
- The annual Sharpe ratio of a strategy without skill over $Y$ years is about $N(0, 1/Y)$
  - Sharpe ratio: mean excess return divided by its standard deviation (Chapter 1); $Y$: the number of years of data
  - mean 0: no skill; variance $1/Y$: the estimation error, which falls with the length of the sample

## The Best of $N$ Strategies (2/2)

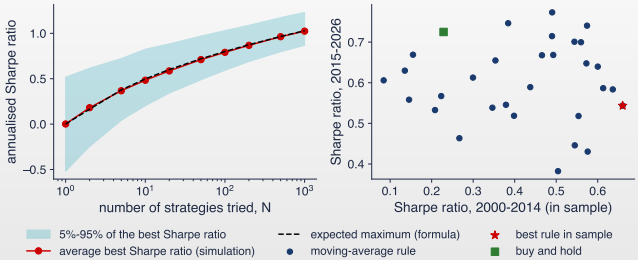
- Expected maximum of $N$ independent zero-skill trials Bailey & López de Prado (2014)

$$\mathrm{E}[\max] \approx \sigma\left[(1-\gamma)\,\Phi^{-1}\left(1-\tfrac{1}{N}\right)+\gamma\,\Phi^{-1}\left(1-\tfrac{1}{Ne}\right)\right]$$

  - $\sigma$: the standard deviation of the Sharpe ratios across trials ($1/\sqrt{Y}$ above); $N$: the number of trials
  - $\Phi^{-1}$: the quantile function of the standard Normal distribution; $e \approx 2.718$; $\gamma = 0.5772$: the Euler–Mascheroni constant
  - $\Phi^{-1}(1-1/N)$ grows roughly like $\sqrt{2\ln N}$: the luck premium rises slowly, but without limit

- Example: 10 years, $\sigma = 1/\sqrt{10} = 0.316$, $N = 100$
  - $0.316\,[(1-\gamma)\,2.326+\gamma\,2.680] = 0.80$: a Sharpe ratio this large is expected from pure luck
  - simulation: the best of 10, 100 and 1000 zero-skill strategies has an average Sharpe ratio of 0.48, 0.79, 1.02

# Selection Bias in Practice




- Left: the best annual Sharpe ratio among $N$ strategies without skill (10 years); right: 30 moving-average rules on the S&P 500 (long or cash), 2000–2014 against 2015–2026
- Interpretation: the best rule in sample (MA 50/200, Sharpe 0.66) earns 0.54 afterwards, rank 22 of 30, below buy and hold (0.72); in and out of sample Sharpe ratios correlate 0.04

SFM_ch13_data_snooping

# The Deflated Sharpe Ratio (1/2)

- Bailey & López de Prado (2014): the probability that the true Sharpe ratio exceeds what the best of $N$ trials would show by luck

$$\mathrm{DSR} = \Phi\Big(\frac{(\widehat{SR} - SR_0)\sqrt{T-1}}{\sqrt{1 - \hat{\gamma}_3 \widehat{SR} + \frac{\hat{\gamma}_4 - 1}{4}\widehat{SR}^2}}\Big)$$

- Notation
  - $\widehat{SR}$: the per-day Sharpe ratio of the chosen strategy; $T$: the number of days
  - $SR_0$: the expected maximum of $N$ zero-skill trials (above), with $\sigma$ = the standard deviation of the $N$ Sharpe ratios tried
  - $\hat{\gamma}_3$: the skewness, $\hat{\gamma}_4$: the kurtosis of the daily returns (3 for the Normal distribution); $\Phi$: the standard Normal distribution function
- Reading
  - the DSR is a probability in $(0, 1)$; above 0.95: the Sharpe ratio is significant after the correction for $N$ trials
  - negative skewness and fat tails enlarge the denominator and lower the DSR

# The Deflated Sharpe Ratio (2/2)

- **Step by step**: the best MA rule, $N = 30$, $T = 3\,769$ days
  - $\widehat{SR} = 0.0416$ per day ($\times\sqrt{252} = 0.66$); $\hat{\gamma}_3 = -0.44$, $\hat{\gamma}_4 = 12.0$
  - $SR_0 = 0.0105\,[(1-\gamma)\,1.834 + \gamma\,2.249] = 0.0219$ (annual 0.35)
  - $z$, the argument of $\Phi$ in the DSR formula: $z = 1.20$, so DSR $= \Phi(z) = 0.88$
  - without the correction for $N$ (PSR, probabilistic Sharpe ratio, $SR_0 = 0$): 0.994
- Interpretation: alone the rule looks significant; after 30 trials it is not (DSR below 0.95), as its later performance confirms

# Empirical Asset Pricing via Machine Learning

- Gu, Kelly & Xiu (2020): which methods best predict the monthly return of individual US stocks?
  - nearly 30,000 stocks, 1957–2016; 94 firm characteristics, interacted with 8 macroeconomic variables, plus 74 industry dummies: 920 features
- Design: 18 years of training (1957–1974), 12 of validation (1975–1986), 30 of test (1987–2016)
  - refit once a year; the training window grows, the validation window rolls forward: a walk-forward design
  - models: OLS, penalised and dimension-reduction regressions, random forest, boosted trees
  - networks NN1–NN5: NN$k$ has $k$ hidden layers



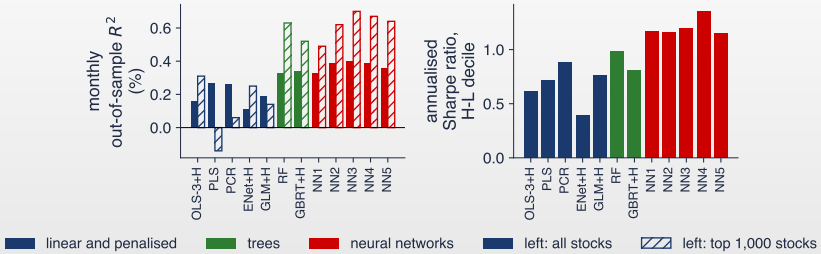
Charles M. Harper Center, Chicago Booth, where Gu and Xiu worked

Photo: Michael Barera (2013); CC BY-SA 4.0; Wikimedia Commons

## Evaluation in the Paper

- $R^2_{OOS} = 1 - \sum_{(i,t)}(r_{i,t+1} - \hat{r}_{i,t+1})^2 / \sum_{(i,t)} r^2_{i,t+1}$: the benchmark is a forecast of **zero**, not the historical mean
  - $r_{i,t+1}$: the return of stock $i$ in month $t+1$; $\hat{r}_{i,t+1}$: its forecast; the sums run over all stock-months of the test period
  - the authors note that the historical mean of a stock is so noisy that it is easy to beat: a zero benchmark is harder
- Diebold–Mariano tests between all pairs of models, with a Bonferroni correction for the 12 comparisons
  - the same tool as Section 7, applied to a panel of stocks
- Economic value: each month, sort stocks into deciles by forecast; buy the top decile, sell the bottom one
  - value-weighted portfolios; Sharpe ratio, drawdown and turnover (the share of the portfolio traded each month)

# Results: Small $R^2$, Large Economic Value




- Left: monthly $R^2_{OOS}$ (Table 1): OLS with all 920 features $-3.46\%$; OLS-3 (size, value, momentum) 0.16%; random forest 0.33%; NN3 0.40% (0.70% for the 1,000 largest stocks)
- Right: annual Sharpe ratio of the value-weighted decile spread (Table 7): OLS-3 0.61, random forest 0.98, NN4 1.35

SFM_ch13_case_gkx

## Interpretation and Limits

- What the paper shows
  - nonlinear models (trees, networks) beat linear ones because they capture interactions between characteristics
  - all methods agree on the main signals: price trends (momentum, short-term reversal), liquidity, volatility
  - shallow networks are enough: 3–4 layers beat 5; "deep" learning does not help with monthly returns
- What to keep in mind
  - an $R^2_{OOS}$ below 1% per month is a large result here: compare with the sign forecasts of Section 8
  - neural portfolios turn over more than 100% of their value each month: trading costs reduce the gains
  - the evidence is a cross-section of thousands of stocks; it says little about timing one index

## An Open Question

- **Do machine-learning volatility forecasts help on the Bucharest Stock Exchange, where liquidity is low and intraday data are scarce?**
  - today: on the S&P 500 the lasso gains +7.1% over HAR; on Bitcoin no significant gain (+2.0%, p 0.520)
  - the BET has no high and low prices in our data, so the proxy must come from squared returns, which are noisier
- Why it is open: thin trading, a different calendar from New York, few crises in the sample
- Related work, as further reading: Pele et al. (2023) (statistical classification of crypto assets); Pele & Mazurencu-Marinescu-Pele (2026) (how precise forecast comparisons can be)
- AI tools can speed up such a study; they do not replace checking it Wang et al. (2023)

## How AI Could Help

- **Literature**: a list of studies on HAR and machine learning for emerging and frontier markets
- **Code**: a first draft of the walk-forward comparison of HAR, lasso and random forest for the BET and three BVB stocks
- **Robustness**: other proxies (absolute returns, two-day returns), other horizons (1, 5, 22 days), S&P 500 features of the previous close
- Example prompt
  - `Write Python code that compares HAR, lasso and a random forest for the weekly log variance of the BET index (proxy: squared daily returns), with yearly walk-forward refits from 2012, purging the last 5 training rows, and reports the out-of-sample R2, QLIKE and Diebold-Mariano tests.`

## What to Check

- Time order: no shuffled folds; features use only data up to the close of day $t$; the last training targets do not reach into the test period
- Scaling and tuning inside the training window only (standardisation, $\lambda$, the number of trees)
- Calendars: the BVB closes before New York; S&P 500 features must come from the previous New York close
- A benchmark in every table (HAR, the majority class, historical simulation) and a test of the difference
- The number of models tried, reported; references: every cited paper must exist; check the DOI

## Project Idea

- **Question**: do machine-learning models forecast the volatility of BVB stocks better than HAR?
  - data: BET, Banca Transilvania, OMV Petrom, BRD (EODHD); S&P 500 and DAX as external features
- Steps
  - a variance proxy for each stock (range where available, squared returns otherwise), weekly target
  - HAR, lasso, random forest and a small MLP, walk-forward from 2014, yearly refits with purging
  - out-of-sample $R^2$, QLIKE and Diebold–Mariano; Shapley values for the external features
- Deliverable: one table, one chart, and a paragraph on what the data can and cannot show
- Declare any AI use, and list the errors of the AI that you corrected

## Key Takeaways

- Machine learning is judged by its error on new data; the bias–variance trade-off sets how flexible a model should be
- Time series need walk-forward validation with purging; random folds on overlapping targets fake predictability
- Ridge and lasso shrink; forests average trees; boosting adds trees; networks learn nonlinear features
- Volatility: HAR is hard to beat, the gains are a few per cent; the sign of returns: no gain over "always up"
- VaR 1%: linear quantile regression works; flexible quantile models lack tail data
- Report the number of trials and deflate the Sharpe ratio; compare every model with a simple benchmark

# Key Formulas

| Quantity | Formula |
|---|---|
| Bias–variance | $E[(y_0 - \hat{f})^2] = (E\hat{f} - f)^2 + \mathrm{Var}(\hat{f}) + \sigma^2$ |
| Ridge | $(X^\top X + \lambda I)^{-1} X^\top y$; 1D: $S_{xy}/(S_{xx} + \lambda)$ |
| Lasso (1D) | $\mathrm{sign}(S_{xy})\max(\lvert S_{xy}\rvert - \lambda/2, 0)/S_{xx}$ |
| Random forest | $\mathrm{Var} = \rho\sigma^2 + (1-\rho)\sigma^2/B$ |
| Boosting | $F_m = F_{m-1} + \nu h_m$ |
| MLP | $\hat{y} = c + \sum_k v_k g(b_k + w_k^\top x)$ |
| HAR | $y_t = \beta_0 + \beta_d \ln v_t + \beta_w \ln v_t^{(w)} + \beta_m \ln v_t^{(m)} + u_t$ |
| $R^2_{OOS}$ | $1 - \sum(y - \hat{y})^2 / \sum(y - \tilde{y})^2$ |
| Quantile loss | $\rho_\alpha(u) = u(\alpha - 1\{u < 0\})$ |
| DSR | $\Phi((\widehat{SR} - SR_0)\sqrt{T-1}/\sqrt{1 - \hat{\gamma}_3 \widehat{SR} + (\hat{\gamma}_4 - 1)\widehat{SR}^2/4})$ |

## Check Yourself

- **Question**: a model has a training $R^2$ of 80% and an out-of-sample $R^2$ of $-5\%$. What happened?
  - **Answer**: overfitting: the model learned the noise of the training sample and loses against the benchmark on new data
- **Question**: why is random K-fold wrong for a 21-day return target?
  - **Answer**: neighbouring targets overlap, so the test answers leak into training; use walk-forward with purging
- **Question**: a classifier is right on 53% of the days and the market rose on 54% of them. Is it useful?
  - **Answer**: no: "always up" is right on 54% of the days
- Next: Chapter 14, crypto assets

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